Advanced Databases: **Project Part 2 – SQL and Accessing a Database**

Title : **Airline Company Database**

Query statements in plain english and sql

-- Names of all passengers who have booked a flight for 25/12.

```
SELECT DISTINCT PNAME
FROM PASSENGER, TICKET, FLIGHT
WHERE FDATE = '2023-12-25' AND TICKET.FID = FLIGHT.FID AND PASSENGER.PSSN = TICKET.PSSN;
```

PNAME
Sofia Pantazidou
Nikolaos Geloudakis

-- IDs of passengers who live in Canada and travel to Spain.

```
SELECT DISTINCT PASSENGER.PSSN
FROM PASSENGER, FLIGHT, TICKET
WHERE PASSENGER.PADDRESS LIKE '%Canada%' AND FLIGHT.FDESTINATION LIKE '%Madrid%' AND
PASSENGER.PSSN = TICKET.PSSN AND TICKET.FID = FLIGHT.FID;
```

PSSN
226262386

-- Find the IDs of airline employees who are over 50 years old.

```
SELECT SSSN
FROM STAFF
WHERE SAGE > 50;
```

SSSN
231054932
835698748
942874598
NULL

-- Find the name of the youngest employee who works as a pilot.

```
SELECT SNAME
FROM STAFF
WHERE SJOBTITLE = 'pilot'
ORDER BY SAGE ASC
LIMIT 1;
```

SNAME
Alexandra Georgopoulou

-- What is the total number of staff that all the airline companies employ?

```
SELECT COUNT(*)
FROM STAFF;
```

COUNT(*)
20

-- Which are the airports that are located in Greece?

```
SELECT ANAME
FROM AIRPORT
WHERE ALOCATION = 'Greece';
```

ANAME
Thessaloniki Airport
Athens Airport

-- Which tickets cost over 300 euros?

```
SELECT TID
FROM TICKET
WHERE TCOST > 300;
```

TID
9
14
15
17
19
20
NULL

-- What is the average ticket cost for every flight?

```
SELECT FLIGHT.FID, AVG(TICKET.TCOST)
FROM FLIGHT, TICKET
WHERE FLIGHT.FID = TICKET.FID
GROUP BY FLIGHT.FID
ORDER BY FLIGHT.FID ASC;
```

FID	AVG(TICKET.TCOST)
1	80.0000
2	150.0000
3	240.0000
4	190.0000
5	300.0000
6	65.0000
7	67.0000
8	50.0000
9	340.0000
10	230.0000
11	300.0000
12	100.0000
13	300.0000
14	2000.0000
15	400.0000
16	700.0000
17	500.0000
18	350.0000
19	150.0000
20	200.0000

-- How many flights will be scheduled during 25/12 – 31/12?

```
SELECT COUNT(*)
FROM FLIGHT
WHERE FDATE BETWEEN '2023-12-25' AND '2023-12-31';
```

	COUNT(*)
▶	2

-- Models of airplanes manufactured after 2010 and have capacity over 100 seats?

```
SELECT PMODEL
FROM AIRPLANE
WHERE PYEAR > 2010 AND PCAPACITY > 100;
```

	PMODEL
▶	Boeing 737
	Boeing 726
	Boeing 777
	Cirrus S222
	Embraer 237
	Boeing 737
	Embraer 195
	Boeing 737

--Delete/Update/Insert SQL Queries

```
DELETE FROM CONNECTS WHERE AID = 9;
DELETE FROM AIRPORT WHERE AID = 9;
DELETE FROM AIRPLANE WHERE PID = 14 AND PYEAR = 1980;

UPDATE STAFF SET SPHONE = 2109999999 WHERE SSSN = 874514254;
UPDATE AIRLINE_COMPANY SET CEMAIL = 'updated@skyairlines.com' WHERE CID = 10;

INSERT INTO FLIGHT VALUES (21, '2023-12-25', 'Madrid Airport', 8);
INSERT INTO PASSENGER VALUES (785468733, 'Ariadni Vogkoli', 2101124658, 'Agios Iwannis 33
Thessaloniki Greece', 35);
```